

Research Paper 05 Summary

Detecting Gas Flares and Estimating Flaring Volumes in Nigeria

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Anejionu et al. (2015), *International Journal of Applied Earth Observation*

1 Basic Information

- **Paper Title:** Detecting Gas Flares and Estimating Flaring Volumes at Individual Flow Stations Using Satellite Data
- **Authors:** Obinna C. D. Anejionu, G. Alan Blackburn, and J. Duncan Whyatt (2015)
- **Focus Region:** Niger Delta, Nigeria (a developing nation environment with heavy oil infrastructure and high vegetation fire background).

2 What is this Paper About? (In Simple Words)

This is the first major research paper that applied satellite fire detection to identify industrial gas flaring in a developing country with complex geography.

- **The Challenge in Developing Countries:** In places like Nigeria (and similarly across India), industrial sites are often surrounded by farmland, forests, and seasonal bushfires. Raw satellite feeds get clogged with thousands of natural fire alerts.
- **The Two-Step Solution:** The authors developed a method combining two simple checks:
 1. **Spatial Proximity (Where is it?):** Checking if the satellite thermal point falls within a specific distance buffer around known oil flow stations, pipelines, and refineries.
 2. **Temporal Persistence (How often does it burn?):** Checking how many times heat is detected at the exact same location over multiple months.

3 Key Concept: The Persistence Rule

The study established that temporary bush fires and crop burning appear at a coordinate for only **1 to 3 days**, whereas operational industrial gas flares produce repeated thermal alerts month after month at the exact same spot.

If a thermal alert occurs near industrial infrastructure AND repeats continuously over a multi-month window, it is classified as a permanent industrial flare with near 100% certainty.

4 How This Helps Our Project (PS26162)

4.1 1. A Direct Blueprint for the Indian Context

India’s industrial zones (such as the oil fields of Assam, the petrochemical hub in Gujarat, and the coal/steel belt in Odisha) face the exact same challenge: heavy seasonal crop fires and dense cloud cover. This paper proves that combining satellite thermal alerts with infrastructure databases works reliably in developing nations.

4.2 2. Methodological Blueprint: Satellite + OpenStreetMap (OSM) Fusion

In our project, we are using OpenStreetMap (OSM) as our facility database. Anejionu et al. provide the blueprint for this exact architecture:

- Ingest NASA FIRMS thermal points.
- Perform a spatial buffer search against OSM industrial polygons.
- If within the buffer, cross-check historical recurrence.

4.3 3. Feature Engineering: The ‘Recurrence Rate’ Feature

We can directly implement their recurrence metric into our machine learning model:

$$\text{Recurrence Rate} = \frac{\text{Number of Active Detections at Coordinate in 90 Days}}{\text{Total Satellite Overpasses in 90 Days}} \quad (1)$$

A high recurrence rate immediately filters out random forest fires and seasonal agricultural burns.

4.4 4. Benchmark for Performance Evaluation

The classification accuracy and flaring site recovery rates published in this paper serve as an established baseline to compare and validate our AI system’s performance.